

SAHIL SANDHU

TECHNICAL PROGRAM MANAGER | TEST SYSTEMS ARCHITECT

EV CHARGING | BATTERY & BMS | HARDWARE-SOFTWARE VALIDATION | IoT ANALYTICS | NPI

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PROFESSIONAL SUMMARY

Technical program and product operations leader with 7 years of progressive experience delivering complex hardware-software products across EV charging, lithium-ion battery/BMS, and telecom power. Owns the complete validation lifecycle - requirements decomposition, test strategy, hardware and software architecture, test-case and acceptance-limit definition, HIL/ATE integration, commissioning, analytics, and production rollout. Led 15+ NPI programs and built automation and IoT platforms supporting 12,000+ deployed units, producing measurable gains in reliability, validation efficiency, MTTR, and time-to-market.

SIGNATURE IMPACT

10+ HIL/ATE systems - 80% less validation effort; 70% faster firmware cycles

12,000+ connected units - 5-10 critical issues detected daily; >40% lower MTTR

45% fewer product failures - 30% higher reliability; three hardware revisions

15+ NPI programs - concept-to-production ownership; 30% faster time-to-market

CORE EXPERTISE

Program & Product: Technical program management, NPI, PRDs, roadmaps, program planning, risk/dependency management, stakeholder, customer and supplier management, KPI/OKR, Agile/Scrum.

Test Systems & Validation: HIL/ATE architecture, validation strategy, test-case and limit definition, hardware/software co-design, firmware validation, system integration, regression/lifecycle testing, FMEA and functional safety.

Software, Data & Protocols: Python, SQL, Pandas, Scikit-learn, Flask, REST APIs, Power BI, predictive analytics, RCA, CAN, UART, RS485, Modbus, BLE and PLCs.

END-TO-END OWNERSHIP

Requirements decomposition | Validation strategy | Test-case and limit definition | Hardware architecture and interface design | Python control software | HIL/ATE integration | Commissioning | Release evidence | Production rollout | Field feedback and continuous improvement

PROFESSIONAL EXPERIENCE

Exicom Tele-Systems Ltd | Gurugram, India

October 2019 - Present

Leading provider of EV charging, energy storage and telecom power solutions.

Deputy Manager - NPI, Projects and Product Operations

April 2024 - Present

- **End-to-end test systems ownership:** Own the architecture and delivery of **10+ Python-automated HIL/ATE systems** for BMS, EVSE and SMR/rectifier platforms - converting product specifications into validation plans; defining test cases, parametric limits and pass/fail logic; designing hardware interfaces and control software; and leading integration, commissioning and production deployment.
- **Validation strategy and release control:** Define product-level test matrices, nominal and boundary test cases, parametric tolerances, acceptance criteria, evidence requirements and release-readiness checkpoints across hardware and firmware revisions.
- **Technical program leadership:** Lead **15+ NPI programs** from concept through mass production, coordinating R&D, firmware, quality, manufacturing, service, suppliers and international customers while driving milestones, technical decisions, risks, dependencies and release readiness.
- **Predictive maintenance at scale:** Architected a production IoT telematics analytics platform for **12,000+ globally deployed charger and battery units**; Python-based anomaly and fault models surface 5-10 high-severity issues daily before customer escalation and reduced MTTR by more than 40%.
- **Product intelligence and governance:** Built Flask/Pandas self-service KPI dashboards to replace spreadsheet and legacy BI workflows; author PRDs, product roadmaps, test protocols, FMEA records, release evidence and program KPIs for engineering and operations decisions.
- **Customer and supplier interface:** Serve as the primary technical liaison for international customers and vendors, translating requirements and field issues into engineering actions, validation evidence, acceptance criteria and closure plans.

Assistant Manager - NPI

April 2023 - April 2024

- **Validation automation:** Engineered Python-based data analysis, firmware flashing and process automation tools that removed engineer-dependent steps from **80% of validation**, shortened firmware cycles by 70% and accelerated product time-to-market by 30%.
- **Reliability improvement:** Led RCA across BMS, battery-pack and laboratory data using Python, Pandas and Scikit-learn; identified systemic defects that drove **three major hardware revisions**, reduced product failures by 45% and improved reliability by 30%.
- **HIL standardization:** Designed automated test rigs and common validation benchmarks for BMS, SMR rectifiers, EV chargers and charge-discharge systems, achieving zero defect escapes across two consecutive product releases.

PROFESSIONAL EXPERIENCE (CONTINUED)

Exicom Tele-Systems Ltd | Gurugram, India

October 2019 - Present

Senior Engineer - NPI

April 2022 - April 2023

- **BMS diagnostics and validation:** Developed Python tools for battery-pack behavior analysis and quantitative SOC/SOH profiling over CAN/UART, improving consistency of fault assessment and reducing field diagnosis time by more than 60%.
- **Engineering analytics:** Built full-stack Flask and Vue.js dashboards that replaced spreadsheet reporting with real-time performance visibility and self-service data exploration for R&D, product and operations stakeholders.
- **Reusable NPI standards:** Created test protocols, FMEA records, engineering checklists and release documentation frameworks that became reusable standards across product lines.

Junior Engineer / Engineer - NPI (Mobility)

July 2020 - April 2022

- **Automation foundation:** Developed battery diagnostic, telemetry logging, behavioral analysis and firmware-flashing tools over CAN and UART, establishing the automation layer later scaled across the validation organization.
- **Hardware development:** Designed and fabricated test jigs, communication interfaces and validation fixtures using CAN, UART and RS485, reducing prototype setup time and supporting faster NPI iteration cycles.
- **Field reliability feedback:** Investigated EV battery-pack failures and translated root-cause findings into design-improvement actions that reduced repeat issues in subsequent product generations.

NPI Engineering Trainee

October 2019 - July 2020

- **Battery telemetry and bench testing:** Built Python scripts for Li-ion pack data logging and field-failure analysis, and supported assembly, wiring and commissioning of battery charge-discharge test setups.
- **Process documentation:** Contributed standardized engineering checklists and NPI technical documentation later adopted as team templates.

SELECTED TECHNICAL PROGRAMS

Automated Multi-Product HIL/ATE Platform

Python | CAN | UART | RS485 | Modbus | PLCs

- Designed modular hardware and software architecture for BMS, EVSE and SMR/rectifier validation, including sequenced tests, firmware operations, data logging, parametric thresholds, exception handling, automated pass/fail decisions and evidence reports.
- Standardized validation benchmarks and traceable release evidence across product lines, enabling parallel, repeatable testing without continuous engineer intervention.

BMS Diagnostic GUI and Real-Time Data Logger [Project demo](#)

Python | PyQt5 | CAN | UART | Pandas

- Created live visualization for cell voltages, temperatures, SOC, SOH, fault flags and charge-discharge curves over CAN/UART for rapid fault interpretation during NPI and field investigations.
- Integrated automated CSV logging and Pandas post-processing, replacing manual oscilloscope-led analysis and reducing field failure diagnosis time by more than 60%.

IoT Predictive Maintenance and Product Analytics

Python | Flask | Pandas | Scikit-learn

- Designed telemetry ingestion, anomaly detection, statistical fault models, service alerts and self-service KPI dashboards that converted field data into preventive-maintenance actions and product-design feedback.
- Operationalized daily fault triage for a 12,000+ unit global install base, surfacing 5-10 high-severity issues each day and enabling service teams to intervene before customer-reported incidents.

Full-Stack Workflow Application

Flask | Vue.js | SQLAlchemy | Redis | Celery

- Built a multi-role application with JWT authentication, REST APIs, relational data models, asynchronous task queues and automated notifications, demonstrating end-to-end software delivery beyond internal engineering tools.

EDUCATION

Indian Institute of Technology Madras (IIT Madras)

2021 - 2025

Bachelor of Science (BS) - Data Science and Applications

Relevant study: machine learning, statistical inference, business analytics, databases, data visualization, deep learning and Python.

INDO-SWISS Training Centre, CSIR-CSIO, Chandigarh

2016 - 2019

Engineering Diploma - Electronics; Mechatronics and Industrial Automation

Relevant study: embedded systems, microcontrollers, industrial PLCs, sensors, PCB design, electronics and control systems.

PUBLICATION

Co-author, "Experiments on Design of Obstacle Avoiding Robots Based on Sensors, Bluetooth, and IoT", Chapter 6, Handbook of Research on the Internet of Things Applications in Robotics and Automation, IGI Global, 2020.

Contribution covered sensor integration, embedded control, Bluetooth/IoT connectivity, real-time data processing and obstacle-avoidance logic for mobile robotics.

TECHNICAL SKILLS

Program & Product: PRDs, roadmaps, program planning, stakeholder/customer/supplier management, risk and dependency management, KPI/OKR, Agile/Scrum and go-to-market coordination.

Test & Systems: HIL/ATE architecture, test strategy, test-case design, acceptance limits, firmware validation, system integration, regression/lifecycle testing, FMEA and functional safety.

Software & Data: Python, SQL, C, Java, Bash, Pandas, NumPy, Scikit-learn, Matplotlib, Flask, Vue.js, PyQt5/PySide2, REST APIs, SQLAlchemy, Redis, Celery, Power BI, Git and JIRA.

Hardware & Protocols: BMS, EVSE/OCPP, SMR/rectifiers, CAN, UART, RS485, Modbus, BLE, I2C, PLCs, relays, embedded systems, firmware flashing, oscilloscopes and logic analyzers.